

Dissection of signaling cascades through gp130 in vivo: reciprocal roles for STAT3- and SHP2-mediated signals in immune responses.

SCIFACT-EVIDENCE-26851674

Evidence abstract

We generated a series of knockin mouse lines, in which the cytokine receptor gp130-dependent STAT3 and/or SHP2 signals were disrupted, by replacing the mouse gp130 gene with human gp130 mutant cDNAs.

The SHP2 signal-deficient mice (gp130F759/F759) were born normal but displayed splenomegaly and lymphadenopathy and an enhanced acute phase reaction.

In contrast, the STAT3 signal-deficient mice (gp130FXQ/FXXQ) died perinatally, like the gp130-deficient mice (gp130D/D).

The gp130F759/F759 mice showed prolonged gp130-induced STAT3 activation, indicating a negative regulatory role for SHP2.

Th1-type cytokine production and IgG2a and IgG2b production were increased in the gp130F759/F759 mice, while they were decreased in the gp130FXXQ/FXXQ immune system.

These results indicate that the balance of positive and negative signals generated through gp130 regulates the immune responses.

Document metadata

Corpus | SciFact

Document ID | 26851674